

Boosting C_{6–12} Higher Alcohols Yield from Ethanol Upgrading over Ca-Deficient Mesoporous Hydroxyapatite through Intermediates Enrichment of C₄ Aldehyde

Jia Wang, Wen-Cui Li, Andi Di, Li-Dong Zhao, Danhui Sun, Lei He, and An-Hui Lu*



Cite This: *ACS Catal.* 2025, 15, 8599–8610



Read Online

ACCESS |



Metrics & More



Article Recommendations



Supporting Information

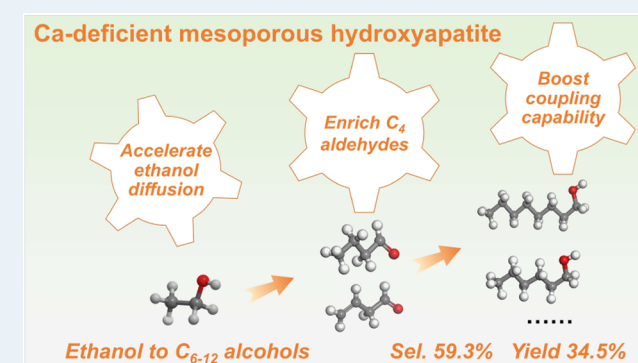
ABSTRACT: Ethanol upgrading to C_{6–12} alcohols offers a green pathway to high-valued chemicals. Achieving high yields of C_{6–12} alcohols remains challenging due to the complexity of the ethanol reaction network. Here, we report an efficient conversion of ethanol to C_{6–12} alcohols using Ca-deficient mesoporous hydroxyapatite as the catalyst, which was prepared from the reaction of NH₃ and PO₄^{3–}. The yield of C_{6–12} alcohols reached 34.5% with a selectivity of 59.3% at 325 °C and 0.1 MPa, exhibiting a high ethanol upgrading performance. Ethanol adsorption kinetics and ethanol-TPD-DRIFT results revealed that the ethanol diffusional time constant of the enlarged pore of the hydroxyapatite increased by 2.01 folds of the original hydroxyapatite, thereby facilitating the access of ethanol molecules to active sites, which was beneficial for ethanol dehydrogenation. The online-generated acetaldehyde was subsequently undergoing aldol condensation to accelerate the formation of C₄ aldehydes, which, as the intermediates, were stabilized over Ca deficiency sites on the hydroxyapatite surface. The enriched C₄ aldehyde intermediates are further coupled with C₂, C₄, or C₆ aldehydes at the basic sites to produce longer-chain aldehydes (C ≥ 6), which are subsequently hydrogenated to form C_{6–12} alcohols. Noticeably, compared to the original hydroxyapatite, the Ca-deficient hydroxyapatite showed enhanced density of basic sites, about two times higher. Hence, this catalyst remarkably boosted the yield of C_{6–12} alcohols from ethanol upgrading. This work provides a strategy for the selective conversion of ethanol to C_{6–12} alcohols.

KEYWORDS: ethanol, C_{6–12} alcohols, hydroxyapatite, C₄ aldehyde, basicity

INTRODUCTION

High-carbon alcohols (C_{6–12} alcohols) are essential feedstock for fine chemicals, extensively used in producing surfactants, plasticizers, and various other chemical products.^{1,2} Meanwhile, C_{6–12} alcohols, as alcohol-based fuels, exhibit combustion values comparable to gasoline with an air-fuel ratio and lower water solubility than low-carbon alcohols, making them attractive candidates for biofuels and oil additives.^{3,4} Currently, the conventional approach for producing high-carbon alcohols is the carbonylation process (oxo process), which uses olefins, carbon monoxide, and hydrogen as feedstock.^{5,6} The conventional process is multistep and energy-intensive relying on fossil energy, making it unsustainable. Therefore, it is necessary to develop a more sustainable process for producing high-carbon alcohols. Ethanol, a widely available biomass-derived platform molecule, presents a promising alternative.^{7–9} Upgrading ethanol through catalytic coupling offers an appealing carbon-neutral approach for producing C_{6–12} alcohols.^{10–12}

In previous studies, solid catalysts containing Lewis acid–base sites such as Mg–Al mixed oxides,^{13,14} metal–supported



catalysts,^{15,16} and hydroxyapatite^{17,18} have been proven effective for upgrading ethanol to high-carbon alcohols. Generally, the ethanol coupling process follows either the “Guerbet” or “direct coupling” pathway. The Guerbet pathway principally involves a series of steps including dehydrogenation, aldol condensation, and hydrogenation.^{19,20} The direct coupling pathway consists of the deprotonation of the β-C of one ethanol molecule, followed by attacking the α-C site of another ethanol molecule for dehydration to produce butanol.^{21,22} The conversion of ethanol to C_{6–12} alcohols proceeds a gradual coupling process of “C₂→C₄→C₆→C₈→C₁₀”, which adapts the stepwise growth model.²³

Received: February 25, 2025

Revised: April 30, 2025

Accepted: May 2, 2025

Published: May 7, 2025



Hydroxyapatite is a promising acid–base catalyst for the upgrading of ethanol to high-carbon alcohols.¹⁷ However, butanol consistently emerges as the predominant product due to the stepwise growth, with reported selectivity exceeding 50% in ethanol coupling reactions catalyzed by hydroxyapatite.^{24,25} The activation and coupling of acetaldehyde and C₄ aldehyde species plays a crucial role in boosting the selectivity of C_{6–12} alcohols. Inspired by these features, the researchers employed transition metals such as Cu, Ni, Co, etc., to enhance the dehydrogenation/hydrogenation capabilities. The improved dehydrogenation capacity facilitates the generation of high concentrations of aldehyde species. However, it is generally necessary to introduce additional hydrogen to allow hydrogenation of butenal and the eventual generation of high-carbon alcohols.²⁶ Otherwise, butenal tends to cyclize to aromatic compounds.²⁷

On the other hand, the development of metal-free loaded hydroxyapatite for the high-yield preparation of C_{6–12} alcohols from ethanol may provide a friendly alternative. According to the ethanol coupling mechanism, achieving a high selectivity for C_{6–12} alcohols depends on the concentration of aldehyde intermediates around the basic sites. This requires the precise design of catalysts with enhanced basicity and the ability to enrich aldehyde intermediates. Particularly, the design of hydroxyapatite with strong basicity and weak acidity remains a great challenge.

In this work, we provide a strategy for boosting C_{6–12} higher alcohols yield from ethanol upgrading using Ca-deficient mesoporous hydroxyapatite. To clarify the role of the active sites and establish a structure–activity relationship, we examined the surface properties of hydroxyapatite using CO₂, NH₃, and butenal as probe molecules. Our investigations revealed that the Ca-deficient hydroxyapatite effectively enriches the C₄ aldehyde intermediates, which subsequently undergo a coupling reaction at the basic sites to generate high-carbon alcohols. Through a combination of kinetic experiments and *in situ* characterization techniques, we propose a reaction mechanism for the conversion of ethanol to C_{6–12} alcohols on the hydroxyapatite.

■ EXPERIMENTAL SECTION

Materials. Monocalcium phosphate (AR, 99%) and calcium hydroxide (AR, 95%) were purchased from Shanghai Aladdin Biochemical Technology Co., Ltd. Diammonium phosphate (AR, 99%), acetaldehyde (AR, 99.5%), ethanol (AR, 99.7%), and butanol (AR, 99.5%) were purchased from Sinopharm Chemical Reagent Co., Ltd. 2-butenal was purchased from ACROS with analytical purity. All of the chemicals were obtained from commercial sources and used without further purification.

Catalyst Preparation. The preparation of hydroxyapatite involved a mechanical ball-milling method. Initially, monocalcium phosphate [Ca(H₂PO₄)₂], calcium hydroxide [Ca(OH)₂], and diammonium phosphate [(NH₄)₂HPO₄] were introduced into the ball-milling tank at a molar ratio of 3:7: (0.05–0.40). Except for special notes, the relative molar ratio of (NH₄)₂HPO₄ for the catalyst preparation was 0.10. The mass ratio of zirconia ball-milling beads of 5 mm diameter to sample mixture was maintained at 20:1. Subsequently, the samples were subjected to ball milling using a Fritsch Pulverisette 7 planetary at 800 rpm for 30 min. After ball milling, the samples were taken out and put in a muffle furnace to calcine at 400 °C for 2 h under static air. The resulting

sample was named (NH₄)₂HPO₄–HAP. For comparison, the contrasting hydroxyapatites were synthesized by introducing NH₃ or H₃PO₄ instead of (NH₄)₂HPO₄ during ball milling and named NH₃–HAP and H₃PO₄–HAP. The size distributions of the powder hydroxyapatite catalysts after ball milling are shown in Figure S1.

Catalyst Characterization. Powder X-ray diffraction (XRD) patterns were obtained using a Panalytical X'pert Pro Super X-ray diffractometer equipped with Cu K α radiation (40 kV, 40 mA, λ = 0.15418 nm). N₂ adsorption–desorption isotherms were recorded at –196 °C by using a Micromeritics TriStar 3000 analyzer. Fourier-transform infrared spectra (FT-IR) were acquired on a Nicolet 6700 spectrometer. Transmission electron microscopy (TEM) images and the high-resolution elemental mapping of energy dispersive X-ray spectroscopy (EDS-mapping) images were obtained using a Tecnai G2 F30 S-Twin instrument (300 kV). NH₃/CO₂-temperature-programmed desorption-mass signals (NH₃/CO₂-TPD-MS) were conducted by using a Micromeritics AutoChem II 2920 apparatus. The time-dependent adsorption profiles of ethanol were collected using an IGA-100 Intelligent Gravimetric Analyzer. The *in situ* Diffuse reflectance infrared Fourier-transform (DRIFT) spectra for CO₂, ethanol, and butenal adsorption were recorded on a Nicolet 6700 spectrometer equipped with a MCT detector and a PIKE DRIFT accessory. The reaction intermediates were captured by a Bruker 70 V spectrometer. Detailed information is provided in the Supporting Information.

Catalytic Activity Measurements. The ethanol upgrading performance was measured at 0.1 MPa in a fixed bed reactor (length 360 mm, inner diameter 8 mm). The system temperature was maintained by a vertically tubular furnace equipped with a YuDian Series 708P controller and a K-type thermocouple. In a typical experiment, 100 mg catalysts were filled in a quartz tube and then pretreated in an N₂ atmosphere at 400 °C for 2 h to remove any H₂O or CO₂ possibly adsorbed. Ethanol was introduced into the reactor via a bubbling method utilizing N₂ as the carrier gas. The pressure of ethanol was regulated by adjusting the flow rate of N₂ and the bubbling temperature. The inlet line is maintained at a constant temperature using a heating band to prevent condensation of the bubbling ethanol. The ethanol (5.7 kPa) diluted with N₂ was passed through the catalyst bed at a total flow rate of 30 mL min^{–1}. The ethanol conversion and product distributions recorded at various temperatures were tested after 0.5 h on the stream, except for special notes. The products were maintained at 240 °C in the gas line before entering the chromatography to avoid condensation of high-carbon substances. The products were quantified using an online Gas Chromatography (GC) with a Flame Ionization Detector (FID). The outlet of the chromatography's six-way valve is connected to a cold trap (0 °C) for collecting the liquid product. The identities of the products were further confirmed using Agilent 7890A GC interfaced with 5975C MS analysis. The retention time was determined by employing the corresponding standard chemicals. For studies of reaction pathways, catalytic performance was measured at 325 °C. Ethanol conversion was varied by changing the catalyst mass from 5 to 200 mg (particle size 40–60 mesh). The bed height was maintained at about 8 mm during the tests with the aid of quartz sand. The test conditions preclude the presence of mass transfer limitations, as supported by our previous work²⁸ and with the Mears criterion and Weisz-Prater criterion (see

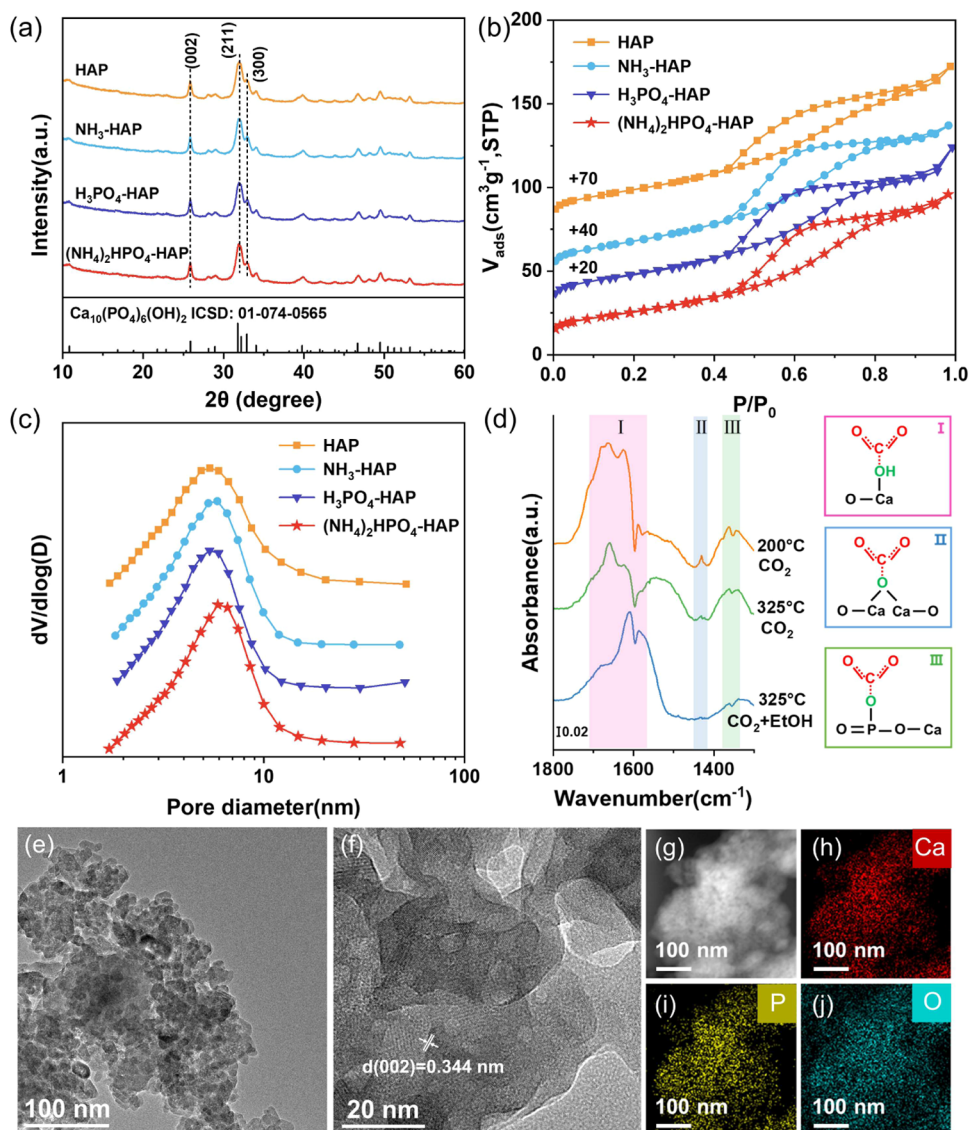


Figure 1. (a) XRD patterns, (b) Nitrogen adsorption–desorption isotherms, and (c) Pore size distributions of the catalysts; (d) FT-IR spectra of (NH₄)₂HPO₄–HAP upon introduction of continuous flow (P_{CO_2} = 5 kPa, at 200 and 325 °C respectively, and P_{CO_2} = 5 kPa and P_{EtOH} = 5 kPa at 325 °C). (e and f) TEM images, (g) HAADF-STEM image, and EDS elemental mapping analysis of (h) Ca, (i) P, and (j) O elemental distributions of (NH₄)₂HPO₄–HAP catalyst.

Supporting Information for details). As for the kinetic test, ethanol conversion and product selectivity were obtained at the initial point (after 0.5 h of reaction on stream). The carbon balance of each sample was better than 95%.

The ethanol conversion and product selectivity were calculated based on moles of carbon in the products, as follows:

Ethanol conversion:

$$\text{con. (\%)} = \left(1 - \frac{n_{\text{out, C}_2\text{H}_5\text{OH}} \times A_{\text{out, C}_2\text{H}_5\text{OH}} \times f_{\text{out, C}_2\text{H}_5\text{OH}}}{n_{\text{out, C}_2\text{H}_5\text{OH}} \times A_{\text{out, C}_2\text{H}_5\text{OH}} \times f_{\text{out, C}_2\text{H}_5\text{OH}} + \sum_{i \geq 2} n_i \times A_i \times f_i} \right) \times 100\%$$

Product selectivity:

$$\text{sel. (\%)} = \frac{n_i \times A_i \times f_i}{\sum_{i \geq 2} n_i \times A_i \times f_i} \times 100\%$$

Ethanol reaction rate:

$$r_{\text{C}_2\text{H}_5\text{OH}} \text{ (mol g}_{\text{cat}}^{-1} \text{ h}^{-1}) = \frac{m_{\text{C}_2\text{H}_5\text{OH}}}{46 \cdot t \cdot m_{\text{cat}}} \times \text{con.}$$

C–C coupling rate:

$$r_{\text{C–C}} \text{ (mol g}_{\text{cat}}^{-1} \text{ h}^{-1}) = r_{\text{C}_2\text{H}_5\text{OH}} \times \sum_{n \geq 2} [\text{sel. C}_{2n} \times (n - 1)]$$

The carbon number, FID peak area, response factor, and molar yield of each product are designated as n_i , A_i , f_i , and m_i , respectively.

RESULTS AND DISCUSSION

The XRD patterns of hydroxyapatite modified with NH₃ and/or H₃PO₄ are similar, displaying characteristic diffraction peaks corresponding to the (002), (211), and (300) crystal planes ascribed to hexagonal hydroxyapatite phase (Figure 1a). The

Table 1. Physical and Chemical Properties of Hydroxyapatite Catalysts

catalysts	S_{BET} ($\text{m}^2 \text{g}^{-1}$)	D_{peak} (nm)	D/r^2 (s^{-1})	basic sites ($\mu\text{mol g}^{-1}$)			acid sites ($\mu\text{mol g}^{-1}$)	basic/acid ratio	Ca/P ratio	
				weak (%)	strong (%)	all			bulk ^a	surface ^b
HAP	103	5.4	3.14×10^{-4}	19.3 (48)	21.0 (52)	40.3	103.7	0.39	1.67	1.48
NH ₃ -HAP	101	6.0	5.86×10^{-4}	17.0 (37)	28.9 (63)	45.9	99.6	0.46	1.67	1.48
H ₃ PO ₄ -HAP	101	5.3	2.64×10^{-4}	15.1 (34)	29.3 (66)	44.4	97.5	0.45	1.64	1.44
(NH ₄) ₂ HPO ₄ -HAP	92	6.3	6.30×10^{-4}	19.2 (25)	57.5 (75)	76.7	88.1	0.87	1.64	1.43

^aBulk means the theoretical Ca/P ratio calculated based on the mass of each reagent used in the preparation. ^bMeasured by XPS.

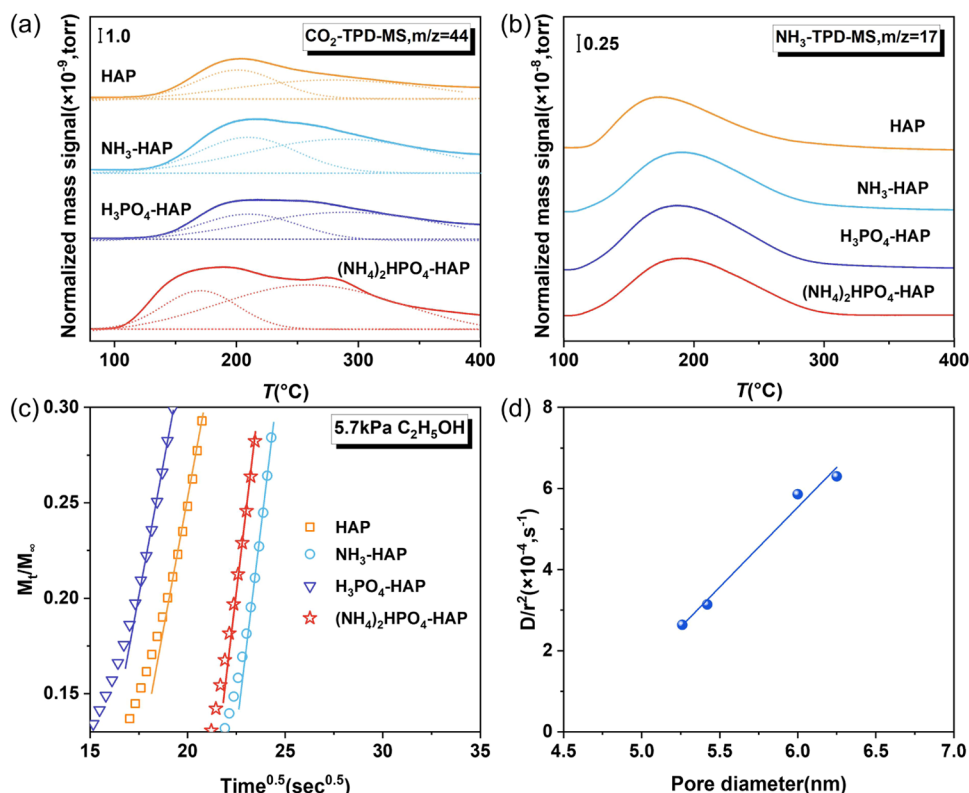


Figure 2. (a) CO₂-TPD-MS signals, (b) NH₃-TPD-MS signals, and (c) Adsorption kinetics profiles (C₂H₅OH, 5.7 kPa) of the catalysts. (d) Correlation between ethanol diffusion time constant and the pore size of each catalyst.

catalysts exhibit low crystallinity with an average crystal size of approximately 12 nm. The N₂ adsorption–desorption isotherms and pore size distribution curves of HAP, NH₃-HAP, H₃PO₄-HAP, and (NH₄)₂HPO₄-HAP samples are illustrated in Figure 1b,c. All four samples exhibit classical type IV isotherms with an H2-type hysteresis loop, indicating the presence of mesopores. Notably, NH₃-HAP and (NH₄)₂HPO₄-HAP show increased pore size compared to HAP (Table 1). This increase is attributed to the scouring effect of ammonia during the formation of the hydroxyapatite structure under mechanical and thermal forces, leading to the formation of large mesopores. Similar enhancements in pore sizes of silica due to ammonia treatment have been documented by Chen et al.²⁹

The surface functional groups and environment of the hydroxyapatite were characterized by FT-IR spectroscopy. The FT-IR spectra of samples displayed three main regions, as shown in Figure S2: the 3500–3800 cm⁻¹ region containing peaks attributed to O–H groups, the 1900–2100 cm⁻¹ region belonging to P=O, and the 1300–1600 cm⁻¹ region derived from carbonate species.^{30,31} The FT-IR spectra of

(NH₄)₂HPO₄-HAP obtained under a continuous flow of CO₂ (5 kPa) at 200 and 325 °C are given in Figure 1d. CO₂ interacts with surface Ca–OH sites to form bicarbonate species and interacts with Ca–O–Ca or Ca–O–P sites to produce carbonate species.^{32,33} At 325 °C, less bicarbonate species were detected, indicating the condensation of some Ca–OH to form Ca–O. Additionally, Figure 1d compared the spectra of (NH₄)₂HPO₄-HAP under a continuous CO₂ feed (5 kPa) with that under continuous CO₂ and ethanol flow. The disappearance of the band at 1420 cm⁻¹ and weakening of the band at 1345 cm⁻¹ indicate ethanol displacing CO₂ from Ca–O–Ca sites and partially from Ca–O–P sites, suggesting that the dehydrogenation of ethanol adsorbed at these sites. Furthermore, the Ca–O–P site serves as the primary active site for C–C coupling, with ethanol also direct coupling at this specific site.²²

The morphology and elemental distribution of the (NH₄)₂HPO₄-HAP sample were analyzed by TEM and energy dispersive spectroscopy (EDS) mapping. Figure 1e,f reveals irregular spherical or cylindrical particles of (NH₄)₂HPO₄-HAP with mesopores of about 10 nm. It

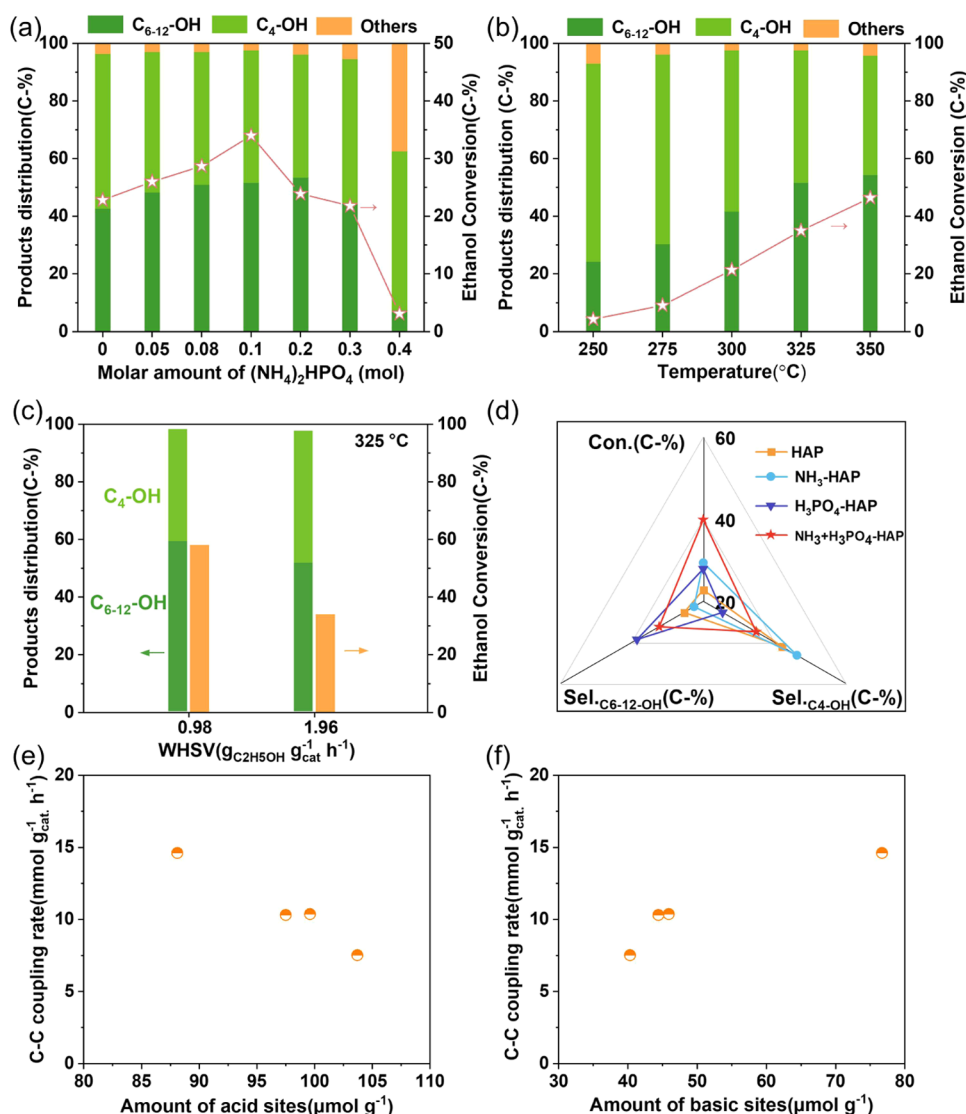


Figure 3. (a) Ethanol conversion and products distribution over the catalysts prepared with different molar amounts of $(\text{NH}_4)_2\text{HPO}_4$ (reaction conditions: 325°C , WHSV of $1.96 \text{ g}_{\text{C}_2\text{H}_5\text{OH}} \text{ g}_{\text{cat}}^{-1} \text{ h}^{-1}$), others include acetaldehyde, ethylene, ether, etc. Ethanol conversion performance over the $(\text{NH}_4)_2\text{HPO}_4$ -HAP catalyst at (b) different temperatures and (c) different WHSV. (d) Ethanol conversion performance over the catalysts (reaction conditions: 325°C , WHSV of $1.96 \text{ g}_{\text{C}_2\text{H}_5\text{OH}} \text{ g}_{\text{cat}}^{-1} \text{ h}^{-1}$). The relationship between the C–C coupling rate and the amount of (e) acid sites and (f) basic sites.

mainly exposes the (002) crystal surface, which is rich in Ca–O–P sites.^{34,35} Compared to H_3PO_4 -HAP (Figure S3), the NH_3 -HAP exhibits a higher presence of mesopores, a trend corroborated by N_2 adsorption–desorption results. STEM-EDS images (Figure 1g–1j) confirmed the uniform distribution of the Ca, O, and P elements on the $(\text{NH}_4)_2\text{HPO}_4$ -HAP samples. There is no elemental N present on the surface, as observed by EDS-mapping of the $(\text{NH}_4)_2\text{HPO}_4$ -HAP-uncalcined sample, suggesting that NH_3 affects only the HAP pore structure and surface properties instead of bonding with Ca, O, and P (Figure S4).

Ethanol is converted to C_{6-12} alcohols over the acid–base catalysts, where a well-balanced acid–base site is vital for achieving higher yields of C_{6-12} alcohols. NH_3 and CO_2 were employed as probe molecules to access the surface acid–base characteristics of hydroxyapatite catalysts. The surface basicity of hydroxyapatite was effectively modulated by the modifications of NH_3 and/or H_3PO_4 , indicated by the CO_2 -TPD-MS curves (Figure 2a). The basic sites on hydroxyapatite are O^{2-}

associated with OH^- and PO_4^{3-} groups. The CO_2 -TPD signals can be divided into two regions: 100 – 200°C , which corresponds to weak basic sites (Ca–OH or Ca–O–Ca), and 200 – 300°C , which is associated with strong basic sites (Ca–O–P).^{22,32} The amounts of basic sites were obtained by integrating from the TCD signals and normalizing them to a unit mass of catalyst (Table 1). The proportion of strong basic sites on HAP was 52%. After modification, the proportion of strong basic sites on $(\text{NH}_4)_2\text{HPO}_4$ -HAP increased to 75%. In addition, the amounts of basic sites were increased to $76.7 \mu\text{mol g}^{-1}$. There are two possible reasons for the enhanced basicity of the modified hydroxyapatite. One is the basic environment provided by NH_3 in hydroxyapatite synthesis.³⁶ The other is the formation of P-rich surfaces via phosphates that leads to Ca deficiency that promotes the improvement of catalyst basicity. On Ca-deficient catalysts, in order to maintain electroneutrality, H^+ was introduced and OH^- was partially replaced by H_2O . Hydrogen bonds were formed between the oxygens of adjoining phosphate groups, thereby enhancing the

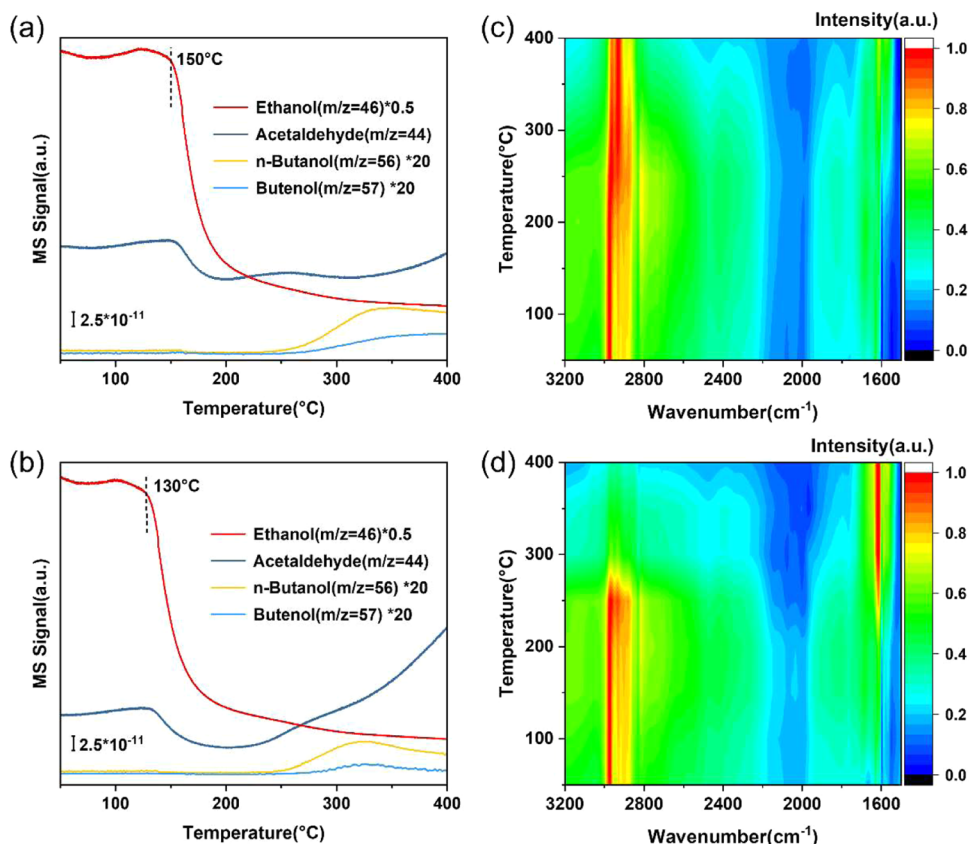


Figure 4. Ethanol-TPD-MS profiles over the (a) HAP and (b) $(\text{NH}_4)_2\text{HPO}_4$ -HAP catalysts. Temperature dependence *in situ* DRIFT spectra for ethanol adsorption on (c) HAP and (d) $(\text{NH}_4)_2\text{HPO}_4$ -HAP catalysts.

basicity.^{18,37} The presence of a Ca-deficient catalyst boosts the formation of $\text{Ca}_V\text{-O-Ca}$ and $\text{Ca}_V\text{-O-P}$ sites, thereby enhancing their ability to adsorb ethanol and product molecules. Additionally, the NH_3 desorption temperatures for all four samples fell within the 100–200 °C range, indicating the presence of weak acid sites on the hydroxyapatite surface (Figure 2b). It is also observed that the acid concentration was clearly reduced after the modification of hydroxyapatite (Table 1).

Adsorption and diffusion of reactants on the catalyst are the initial steps in catalytic reactions. We measured the time-resolved gas adsorption of ethanol on hydroxyapatites to investigate the adsorption behavior of ethanol (Figure 2c, Table 1). Surprisingly, the diffusional time constant ($D' = D/r^2$, s^{-1}) of NH_3 -modified catalysts, such as NH_3 -HAP and $(\text{NH}_4)_2\text{HPO}_4$ -HAP, were 5.86×10^{-4} and 6.30×10^{-4} , respectively, which were 1.87 and 2.01 folds of the HAP. As anticipated, the D' is positively correlated with pore size, as shown in Figure 2d. This implies that the catalyst's pore size increased with the modification of NH_3 , facilitating the adsorption and diffusion of ethanol molecules. The enhanced collision efficiency between ethanol molecules and active sites accelerates the reaction rate. Furthermore, hydroxyapatite with enlarged pore size can also facilitate the diffusion of products such as butanol (Figure S5).

The catalytic activity of the hydroxyapatite modified with varying molar amounts of $(\text{NH}_4)_2\text{HPO}_4$ is illustrated in Figure 3a and Tables S1, S2. Increasing the relative molar amount of $(\text{NH}_4)_2\text{HPO}_4$ to 0.1 mol enhanced the ethanol conversion from 22.8 to 35.0% and improved the selectivity toward C_{6-12}

alcohols from 42.7 to 51.7%. The C_{12+} alcohols were not detected in the product. Aldehyde species, such as acetaldehyde and butanal, etc., have a total selectivity of less than 2% under reaction conditions. It was demonstrated that these aldehyde species were sufficiently hydrogenated and coupled during the reaction to form the target product, high-carbon alcohols. Whereas, when the amount of $(\text{NH}_4)_2\text{HPO}_4$ was further increased, a significant decrease in the catalyst activity occurred, likely due to the reduction in the basicity of the catalyst surface caused by excess phosphate (Table S3). By optimization, we chose the catalyst synthesized using a $(\text{NH}_4)_2\text{HPO}_4$ molar ratio of 0.1 mol as a representative sample for further analyses. Initially, a sharp increase in the ethanol conversion was observed with an increase in reaction temperature, accompanied by an enhancement in the selectivity of C_{6-12} alcohols from 24.3% at 250 °C to 54.4% at 350 °C (Figure 3b). It was noted that the byproducts selectivity became higher at 350 °C. Therefore, 325 °C was selected as the optimum reaction temperature for the subsequent analyses. Additionally, by reducing the weight hour space velocity (WHSV) to $0.98 \text{ g}_{\text{C}_2\text{H}_5\text{OH}} \text{ g}_{\text{cat}}^{-1} \text{ h}^{-1}$, the same catalyst achieved a remarkable ethanol conversion of 58.1% and an impressive selectivity of 59.3% toward C_{6-12} alcohols (Figure 3c). A high yield of C_{6-12} alcohols (34.5%) was obtained on $(\text{NH}_4)_2\text{HPO}_4$ -HAP in the absence of transition metal copper, which is comparable to the performance of the reported Cu-HAP catalyst (39.3% yield of C_{6-12} alcohols), as shown in Table S4. Compared to other reported solid acid–base catalysts, our catalysts exhibit significant advantages in both high selectivity and high yield for the

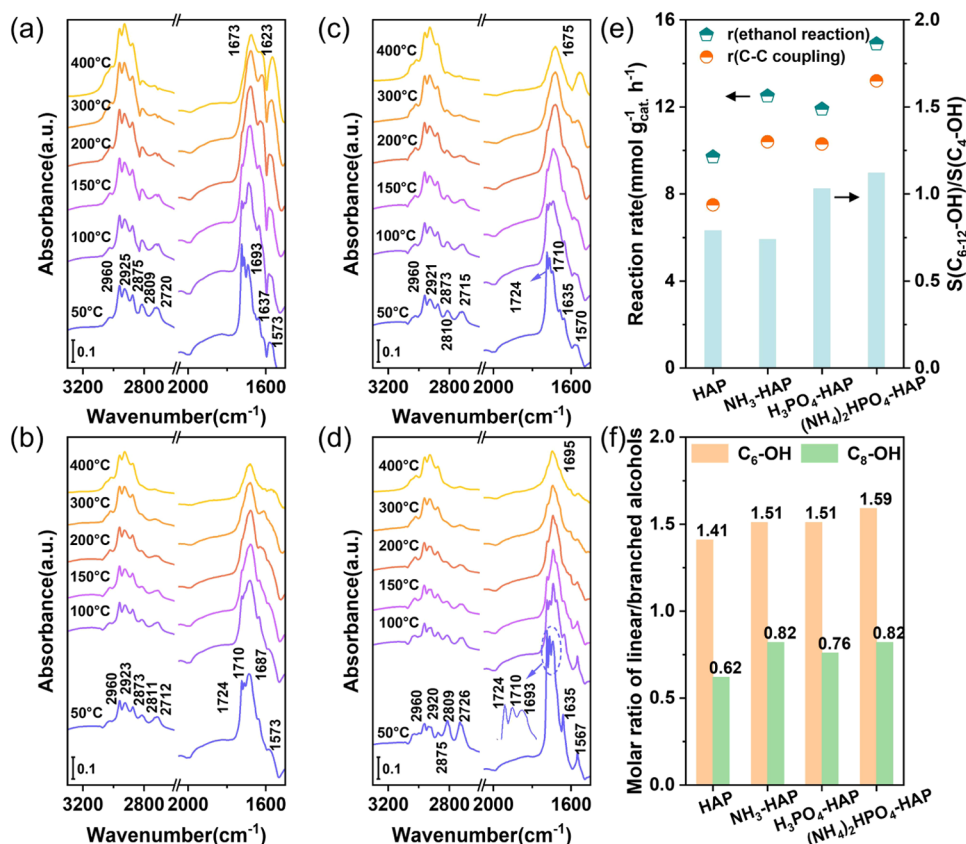


Figure 5. Temperature dependence *in situ* DRIFT spectra for butenal adsorption over (a) HAP, (b) NH₃-HAP, (c) H₃PO₄-HAP, and (d) (NH₄)₂HPO₄-HAP catalysts. (e) Tendency of ethanol reaction rate, C-C coupling rate, and $S(\text{C}_{6-12}\text{-OH})/S(\text{C}_4\text{-OH})$ over the catalysts at 325 °C with a WHSV of 1.96 g_{C₂H₅OH} g_{cat}⁻¹ h⁻¹. (f) Molar ratio of linear and branched alcohols in the product over the catalysts at 325 °C with a WHSV of 1.96 g_{C₂H₅OH} g_{cat}⁻¹ h⁻¹.

upgrading of ethanol to C₆₋₁₂ alcohols. The (NH₄)₂HPO₄-HAP catalyst exhibited good regeneration stability in ethanol conversion (Figure S6).

As shown in Figure 3d, the ethanol conversion on NH₃-HAP increased from 22.8 to 29.4% compared to HAP, indicating that the scouring effect of NH₃ led to the formation of enlarged mesopores, which promoted the diffusion of ethanol molecules and thus facilitated the ethanol conversion. For the H₃PO₄-HAP catalyst, the selectivity of C₆₋₁₂ alcohols was improved from 42.7 to 49.3%. The charge imbalance induced by PO₄³⁻ resulted in the formation of Ca-deficient sites, which enhanced the basicity and thus exhibited an improved C-C coupling ability. The NH₃+H₃PO₄-HAP catalyst showed ethanol conversion of 35.0% and C₆₋₁₂ alcohols selectivity of 51.7%, with performance similar to that of (NH₄)₂HPO₄-HAP (Figure S7), suggesting that Ca-deficient mesoporous hydroxyapatite prepared by the action of NH₃ and H₃PO₄ can efficiently accelerate ethanol dehydrogenation and C-C coupling reaction.

Further experiments were designed to investigate the specific steps at which (NH₄)₂HPO₄ functions. Figure S8 exhibits the effect of the addition of (NH₄)₂HPO₄ at different stages of hydroxyapatite synthesis on the catalyst performance. Among them, (I) denotes coball milling of Ca(H₂PO₄)₂, Ca(OH)₂, and (NH₄)₂HPO₄ for 30 min, and (II) denotes ball milling of Ca(H₂PO₄)₂ and Ca(OH)₂ for 15 min followed by coball milling of (NH₄)₂HPO₄ for 15 min. These two samples showed identical performance. Notably, the same crystalline structure can be formed in Ca(H₂PO₄)₂ and Ca(OH)₂ ball

milled for 15 min as ball milled for 30 min (Figure S9). (III) denotes the ball milling of Ca(H₂PO₄)₂ and Ca(OH)₂ for 30 min, followed by physical mixing with (NH₄)₂HPO₄. Sample (III) showed a significant decrease in the ethanol reaction. The results demonstrated that NH₃ and PO₄³⁻ decomposed by (NH₄)₂HPO₄ during ball milling influenced the surface properties and pore structure of hydroxyapatite after the formation of its crystalline phase.

We tried to correlate the C-C coupling rate with the amount of basic sites, acid sites, and the basic/acid ratio of the catalysts (Figures 3e,f and Figure S10). In general, the results suggested that the C-C coupling rate increased with the amount of basic sites and basic/acid ratio while it decreased with the amount of acid sites. This indicates that the basic sites are the essential active sites responsible for C-C coupling.

To further elucidate the role of NH₃ and H₃PO₄ in the modification of hydroxyapatite and the underlying reaction mechanism, we conducted ethanol temperature-programmed desorption-mass spectrometry experiments (ethanol-TPD-MS) on each catalyst. The activation temperature of ethanol on the HAP was observed to be 150 °C, as shown in Figure 4a, with the acetaldehyde product detectable at this temperature. In the MS spectra of the other three modified hydroxyapatite samples, ethanol activation, and acetaldehyde production were noted in the lower temperature region (Figure 4b and S11). This illustrates that at low reaction temperatures, ethanol dehydrogenation is the initial step in generating high-carbon alcohols.³⁸ Acetaldehyde signals were significantly enhanced on the NH₃-modified catalysts, indicating accelerated diffusion

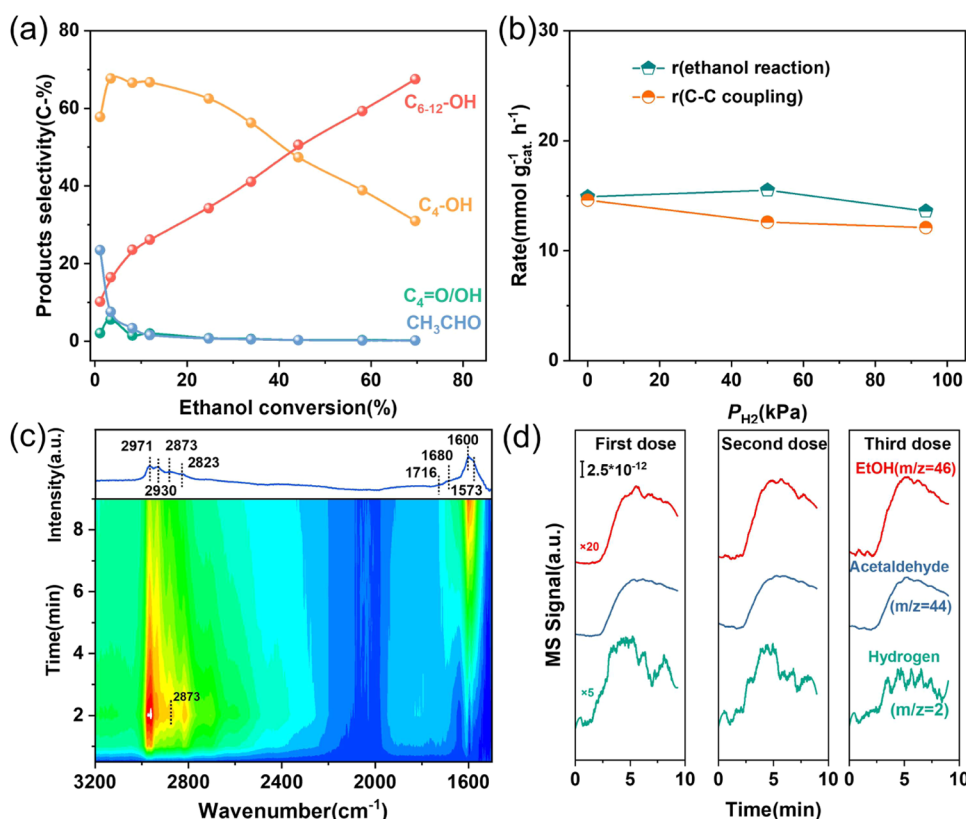


Figure 6. Performance and characterizations of the $(\text{NH}_4)_2\text{HPO}_4$ -HAP catalyst. (a) Dependence of the product distributions as a function of the ethanol conversion (325 °C, WHSV of 0.98–40 $\text{g}_{\text{C}_2\text{H}_5\text{OH}} \text{g}_{\text{cat}}^{-1} \text{h}^{-1}$). (b) Dependence of the ethanol reaction rate and C–C coupling rate on the H_2 partial pressure (325 °C, WHSV of 1.96 $\text{g}_{\text{C}_2\text{H}_5\text{OH}} \text{g}_{\text{cat}}^{-1} \text{h}^{-1}$). (c) *In situ* DRIFT spectra recorded at 325 °C after one dose of ethanol (5×10^{-8} mol) using ~ 40 mg of catalyst. (d) MS signals after each dose of ethanol (5×10^{-8} mol).

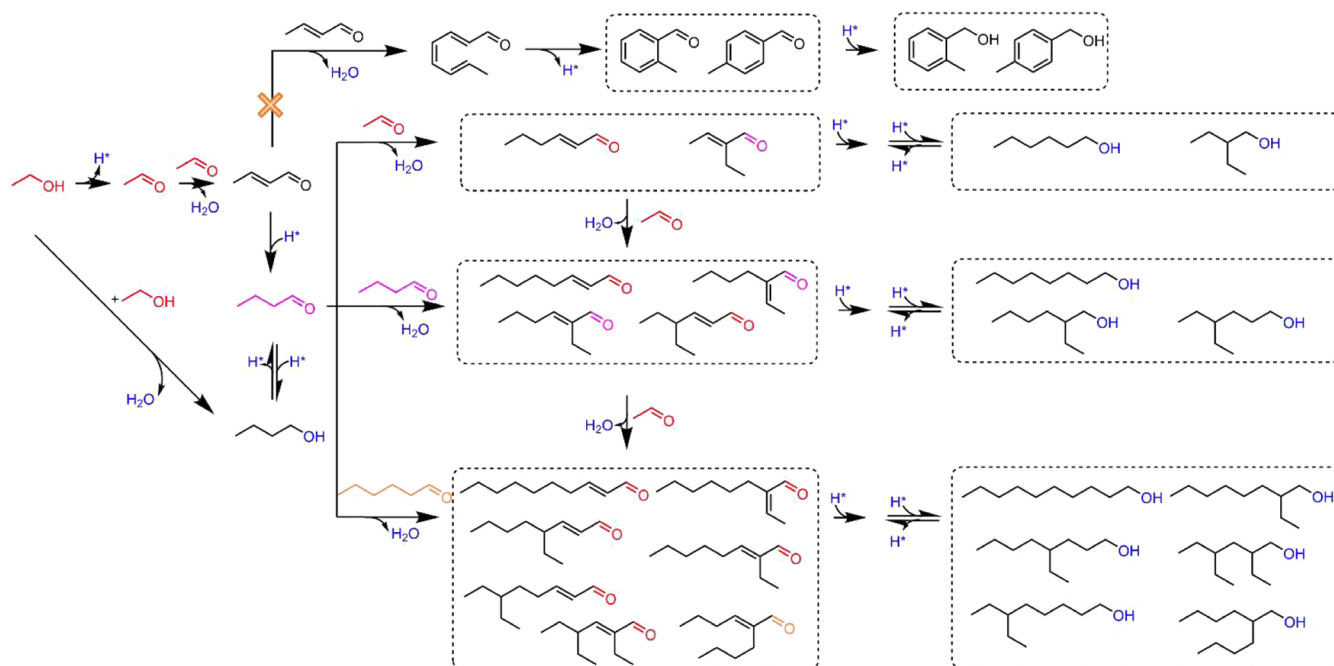
of ethanol molecules on NH_3 -HAP and $(\text{NH}_4)_2\text{HPO}_4$ -HAP, thereby facilitating access of ethanol to the active sites, which was beneficial for accelerating ethanol dehydrogenation, consistent with the previous findings. The generation of the coupling products butenol and butanol was observed at desorption temperatures above 250 °C. Only trace amounts of butanal and butenal were recorded, suggesting rapid hydrogenation of the carbonyl group on these species.

To gain additional insights into the intermediates during ethanol desorption, we conducted temperature-dependent *in situ* DRIFT spectra for ethanol adsorption on hydroxyapatite catalysts (Figure 4c,d, and S11). Among them, the bands at 2972, 2927, and 2897 cm^{-1} are attributed to the stretching modes of CH_3 and CH_2 groups that adsorb ethoxylates,^{39,40} and the band at 2870 cm^{-1} belongs to butanol (Figure S12).²² The band at 1710 cm^{-1} is assigned to the $\nu(\text{C}=\text{O})$ of acetaldehyde.¹² The peaks at 1670 and 1640 cm^{-1} are attributed to $\nu(\text{C}=\text{O})$ and $\nu(\text{C}=\text{C})$ of butenal or butenol species.^{1,16} As the temperature increased, the intensity of the peaks corresponding to ethanol adsorption decreased, suggesting the desorption of ethanol and its conversion into other compounds. The butanol signal on H_3PO_4 -HAP was observed at 100 °C, while the intensity of peaks corresponding to acetaldehyde and butenal gradually increased above 200 °C. For the H_3PO_4 -modified catalysts, the peaks at 2972, 2927, and 2897 cm^{-1} corresponding to ethoxylates were weakened by increasing the temperature to 300 °C, indicating the strong ethanol upgrading ability of H_3PO_4 -HAP and $(\text{NH}_4)_2\text{HPO}_4$ -HAP.

Furthermore, we investigated the adsorption properties of various catalysts on C_4 aldehyde intermediates by using butenal-TPD-DRIFT analysis. As shown in Figures 5a–5d, the bands at 1710 cm^{-1} , 2720/2810 cm^{-1} , and 2873/2960 cm^{-1} belong to $\nu(\text{C}=\text{O})$, aldehydic $\nu(\text{C}-\text{H})$, and aliphatic $\nu(\text{C}-\text{H})$ for adsorbed butenal at 50 °C, respectively.^{1,16} The aldehydic $\nu(\text{C}-\text{H})$ band was still observed at 300 °C on NH_3 -HAP, implying a more stable adsorption of butenal over the catalyst surface. Surface modification with NH_3 effectively enhances the stability of butenal adsorption on the hydroxyapatite surface, promoting its enrichment. The band at 1570 cm^{-1} is attributed to aromatic species and is probably generated by overcondensation of aldehyde species.²⁶ H_3PO_4 -modified catalysts significantly suppressed the overcondensation of butenal to aromatic compounds, as evidenced by the diminished peak intensity in the 1570 cm^{-1} band. Thus, the synergistic modification of NH_3 and H_3PO_4 can facilitate more stable adsorption of butenal on hydroxyapatite. To conclude, enriching the C_4 aldehyde and promoting aldol condensation to foster linear carbon chain growth is crucial for the highly selective preparation of C_{6-12} alcohols. Higher $S(\text{C}_{6-12}-\text{OH})/S(\text{C}_4-\text{OH})$ ratios were achieved with $(\text{NH}_4)_2\text{HPO}_4$ -HAP (Figure 5e).

Moreover, we have measured the hexenal-TPD-DRIFT characterization, as shown in Figure S13. The $\nu(\text{C}-\text{H})$ band (2726 cm^{-1}) of the aldehyde on $(\text{NH}_4)_2\text{HPO}_4$ -HAP is still observed above 150 °C, whereas it is almost invisible at 100 °C on HAP. It indicated that $(\text{NH}_4)_2\text{HPO}_4$ -HAP has more stable adsorption of hexenal (C_6 aldehyde) compared to HAP. This is further evidenced by the higher selectivity of C_{8+} alcohols

Scheme 1. Proposed Mechanism of the Catalytic Conversion of Ethanol to High Alcohols over the $(\text{NH}_4)_2\text{HPO}_4\text{-HAP}$ Catalyst



observed for $(\text{NH}_4)_2\text{HPO}_4\text{-HAP}$, as presented in Table S1. Ca-deficient hydroxyapatite possesses a negatively charged surface, which facilitates the adsorption of partially positively charged aldehyde species through electrostatic interactions and hydrogen bonding. Therefore, $(\text{NH}_4)_2\text{HPO}_4\text{-HAP}$ can effectively enrich aldehyde species. Indeed, for the highly selective generation of high-carbon alcohols, the enrichment of C_2 , C_4 , and C_6 aldehydes is essential. Given the relatively high selectivity of butanol (C_4 alcohol) in previous studies, in this work, we emphasized the importance of the enrichment of C_4 aldehydes for the generation of highly selective C_{6-12} alcohols.

The ethanol reaction rate and C–C coupling rate of the catalyst were determined to provide kinetic insights (Figure 5e). The HAP displayed a lower reaction rate, whereas the ethanol and C–C coupling rates on $(\text{NH}_4)_2\text{HPO}_4\text{-HAP}$ were enhanced 1.5-fold and 1.8-fold, respectively. The results obtained are consistent with the *in situ* DRIFT results. Also, the yield of C_{6-12} alcohols on the $(\text{NH}_4)_2\text{HPO}_4\text{-HAP}$ catalyst was increased to almost two times that on HAP (Figure S14). On the other hand, the aldol condensation proceeds via a nucleophilic attack mechanism, where the rate-determining step involves deprotonation of the nucleophile aldehyde at the basic site.^{23,41} The deprotonated aldehyde is coupled to the electrophilic aldehyde via aldol condensation. An aldol condensation involving two aldehydes can yield one of two possible products, with the linear or branched structure of the product being determined by which aldehyde undergoes deprotonation before nucleophilic attack. The addition reaction involving C_2 aldehyde deprotonation with C_4 and C_6 aldehydes leads to the formation of linear aldehydes, as depicted in Figure S15. Conversely, deprotonation of C_4 and C_6 aldehydes when reacting with C_2 aldehydes forms branched aldehydes. The $\alpha\text{-H}$ pKa of aldehydes increases as the carbon chain grows, making deprotonation of aldehydes species quite challenging.^{11,30} The enhanced molar ratio of the linear/branched alcohols on the $(\text{NH}_4)_2\text{HPO}_4\text{-HAP}$ catalyst

indicates improved dehydrogenation and coupling capacities, facilitating rapid deprotonation of generated acetaldehyde for subsequent aldol condensation with enriched C_4 aldehyde, followed by hydrogenation to produce linear alcohols (Figure 5f).

The reaction pathways on $(\text{NH}_4)_2\text{HPO}_4\text{-HAP}$ were elucidated through reaction space-time study and *in situ* DRIFT. As shown in Figure 6a, selective extrapolation to near-zero conversion indicates that both acetaldehyde and butanol are the primary products. The selectivity toward butanol exhibits an initial increase followed by a subsequent decrease with ethanol conversion, implying its role as a secondary product. In contrast, the selectivity toward butenal and butanol in the products is notably lower than that of butanol. The selectivity of the C_{6-12} alcohols improved with increasing ethanol conversion. The product distributions indicate that the ethanol coupling process follows a step-growth model (Figure S16).

We further investigated the effect of the H_2 partial pressure on the reaction rate. The ethanol reaction rate and the C–C coupling rate showed no dependence on H_2 partial pressure, as illustrated in Figures 6b and S17. Consistent with previous reports,³² hydrogenation during the process likely originates from ethanol through the hydrogen transfer mechanism. Furthermore, we conducted *in situ* DRIFT experiments to analyze the species present on the catalyst surface after pulsed ethanol. The observation revealed a distinct peak in butanol (2873 cm^{-1}) levels occurring within 2 min following a single pulse of ethanol, confirming the direct conversion of ethanol into butanol (Figure 6c). The $\nu(\text{C}=\text{O})$ band (1710 cm^{-1}) of acetaldehyde gradually intensified after 4 min. Supported by the mass spectrometry data (Figure 6d), the signal associated with acetaldehyde was detected after each ethanol pulse. These findings collectively demonstrate that ethanol coupling on $(\text{NH}_4)_2\text{HPO}_4\text{-HAP}$ catalysts proceeds via both the Guerbet and direct coupling pathways.

The conclusive agreement between *in situ* DRIFT and ethanol-TPD further confirms the reaction pathway for the conversion of ethanol to high-carbon alcohols over the $(\text{NH}_4)_2\text{HPO}_4$ –HAP catalyst. Based on these findings, a reaction mechanism for ethanol coupling was proposed (Scheme 1). Ethanol adsorbs on the Ca_V –O–Ca and Ca_V –O–P sites of hydroxyapatite and undergoes dehydrogenation at the Ca–O site, producing acetaldehyde molecules that migrate to the Ca–O–P site for aldol condensation to form butenal. Subsequent hydrogenation converts butenal to butanal. Selective hydrogenation of butanal yields butanol, or it can further condense with C_2 , C_4 , or C_6 aldehydes to produce longer carbon chain aldehydes ($\text{C} \geq 6$), which are subsequently hydrogenated to form high-carbon alcohols. Alternatively, ethanol directly couples at the Ca–O–P site to form butanol, which is then dehydrogenated to form C_{6-12} alcohols through self-coupling or cross-coupling.

CONCLUSIONS

In summary, we provide a strategy for developing a Ca-deficient mesoporous hydroxyapatite for enhancing the performance of ethanol conversion to C_{6-12} alcohols. The $(\text{NH}_4)_2\text{HPO}_4$ –HAP achieved a 59.3% selectivity toward C_{6-12} alcohols with an ethanol conversion of 58.1% at 325 °C with a WHSV of 0.98 $\text{g}_{\text{C}_2\text{H}_5\text{OH}} \text{g}_{\text{cat}}^{-1} \text{h}^{-1}$, surpassing the most reported catalysts. The NH_3 -modified hydroxyapatite enhanced the ethanol diffusion rate, improving the collision efficiency between ethanol and active sites, which was beneficial for ethanol dehydrogenation. The H_3PO_4 -modified hydroxyapatite surface induced Ca deficiency, generating Ca_V –O–Ca and Ca_V –O–P sites that promoted the adsorption of ethanol and product molecules. *In situ* DRIFT and probe adsorption reveal that these active sites can stabilize the C_4 aldehyde intermediates, facilitating their enrichment. The synergistic modification of hydroxyapatite with NH_3 and H_3PO_4 effectively enhances the ethanol dehydrogenation and subsequent coupling process, thereby driving the reaction toward increased ethanol conversion and selectivity to high-carbon alcohols. Ethanol is converted to C_{6-12} alcohols via both the Guerbet and direct coupling pathways. This study introduces new possibilities for the precise design of efficient catalysts for ethanol upgrading.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acscatal.5c01449>.

Characterization methods; mass transfer calculations; particle size distributions, FT-IR spectra, TEM images, EDS element mapping images of the hydroxyapatite catalysts; adsorption kinetics profiles for butanol; stability and regeneration tests of $(\text{NH}_4)_2\text{HPO}_4$ –HAP catalyst; performance of NH_3 + H_3PO_4 –HAP catalyst; performance under different $(\text{NH}_4)_2\text{HPO}_4$ addition methods and on different H_2 partial pressure; XRD pattern of sample; relationship between the C–C coupling rate and basic/acid ratio; ethanol-TPD-MS profiles and *in situ* DRIFT spectra on hydroxyapatite catalysts; FT-IR spectra of aldehyde and alcohol species; *in situ* DRIFT spectra for hexenal adsorption; yield of alcohols of hydroxyapatite catalysts; formation pathways of linear and branched alcohols; step-growth probability;

GC profile of products; performance of ethanol coupling to high alcohols, distribution of major products, summary of catalytic activity over the catalysts; and physical and chemical properties of hydroxyapatite catalysts (PDF)

AUTHOR INFORMATION

Corresponding Author

An-Hui Lu – State Key Laboratory of Fine Chemicals, Liaoning Key Laboratory for Catalytic Conversion of Carbon Resources, School of Chemical Engineering, Dalian University of Technology, Dalian 116024, P. R. China; Liaoning Binhai Laboratory, Dalian 116023, P. R. China; orcid.org/0000-0003-1294-5928; Phone: +86-0411-84986112; Email: anhuilu@dlut.edu.cn.

Authors

Jia Wang – State Key Laboratory of Fine Chemicals, Liaoning Key Laboratory for Catalytic Conversion of Carbon Resources, School of Chemical Engineering, Dalian University of Technology, Dalian 116024, P. R. China

Wen-Cui Li – State Key Laboratory of Fine Chemicals, Liaoning Key Laboratory for Catalytic Conversion of Carbon Resources, School of Chemical Engineering, Dalian University of Technology, Dalian 116024, P. R. China; orcid.org/0000-0001-8066-7144

Andi Di – State Key Laboratory of Fine Chemicals, Liaoning Key Laboratory for Catalytic Conversion of Carbon Resources, School of Chemical Engineering, Dalian University of Technology, Dalian 116024, P. R. China

Li-Dong Zhao – State Key Laboratory of Fine Chemicals, Liaoning Key Laboratory for Catalytic Conversion of Carbon Resources, School of Chemical Engineering, Dalian University of Technology, Dalian 116024, P. R. China

Danhui Sun – State Key Laboratory of Fine Chemicals, Liaoning Key Laboratory for Catalytic Conversion of Carbon Resources, School of Chemical Engineering, Dalian University of Technology, Dalian 116024, P. R. China

Lei He – State Key Laboratory of Fine Chemicals, Liaoning Key Laboratory for Catalytic Conversion of Carbon Resources, School of Chemical Engineering, Dalian University of Technology, Dalian 116024, P. R. China

Complete contact information is available at: <https://pubs.acs.org/10.1021/acscatal.5c01449>

Author Contributions

A.-H.L. conceived and designed the project. J.W. conducted the catalyst preparation, catalytic experiments, characterization, and spectroscopic investigation; L.-D.Z. and D.S. assisted in performing kinetic measurements. W.-C.L., A.D., L.H., and A.-H.L. have discussed the results and revised the manuscript. All authors have reviewed and contributed to the published version of the manuscript.

Notes

The authors declare no competing financial interest.

ACKNOWLEDGMENTS

This work was financially supported the Key Program Project of the Joint Fund of National Natural Science Foundation of China (U23A20130), and the Liaoning Binhai Laboratory (LBLE-2023-04).

REFERENCES

- (1) Yuan, L.; Zhang, M.; Fan, G.; Li, F. Remarkably Boosting Ethanol Upgrading to Higher Alcohols through Cooperative Catalysis of Surface-Interface Multiple-Active Sites on Nitrogen-Doped Carbon Decorated Copper-Based Catalysts. *Appl. Catal., B* **2024**, *343*, No. 123488.
- (2) Lin, X.; Fei, X.; Chen, D.; Qi, Y.; Xu, Q.; Liu, Y.; Zhang, Q.; Li, S.; Wang, T.; Qin, Y.; Qiu, X. Efficient Catalytic Upgrading of Ethanol to Higher Alcohols via Inhibiting C–C Cleavage and Promoting C–C Coupling over Biomass-Derived NiZn@NC Catalysts. *ACS Catal.* **2022**, *12* (19), 11573–11585.
- (3) Gagliardi, A.; Balestra, G.; De Maron, J.; Mazzoni, R.; Tabanelli, T.; Cavani, F. Ethanol to Gasoline and Sustainable Aviation Fuel Precursors: An Innovative Cascade Strategy over Zr-based Multifunctional Catalysts in the Gas Phase. *Appl. Catal. B-Environ. Energy* **2024**, *349*, No. 123865.
- (4) Xue, M.; Yang, B.; Xia, C.; Zhu, G. Upgrading Ethanol to Higher Alcohols via Biomass-Derived Ni/Bio-Apatite. *ACS Sustainable Chem. Eng.* **2022**, *10* (11), 3466–3476.
- (5) Landis, C. R. Construction and Deconstruction of Aldehydes by Transfer Hydroformylation. *Science* **2015**, *347* (6217), 29–30.
- (6) Li, Z.; Luo, G.; Hu, Z.; Pei, X.; Zeng, Z.; Guo, S.; Lv, J.; Huang, S.; Wang, Y.; Ma, X. Zinc Blende CoO as an Efficient CO Nondissociative Adsorption Site for Direct Synthesis of Higher Alcohols from Syngas. *ACS Catal.* **2024**, *14* (4), 2181–2193.
- (7) Purdy, S. C.; Collinge, G.; Zhang, J.; Borate, S. N.; Unocic, K. A.; Wu, Q.; Wegener, E. C.; Kropf, A. J.; Samad, N. R.; Yuk, S. F.; Zhang, D.; Habas, S.; Krause, T. R.; Harris, J. W.; Lee, M.-S.; Glezakou, V.-A.; Rousseau, R.; Sutton, A. D.; Li, Z. Dynamic Copper Site Redispersion through Atom Trapping in Zeolite Defects. *J. Am. Chem. Soc.* **2024**, *146* (12), 8280–8297.
- (8) Hanukovich, S.; Dang, A.; Christopher, P. Influence of Metal Oxide Support Acid Sites on Cu-Catalyzed Nonoxidative Dehydrogenation of Ethanol to Acetaldehyde. *ACS Catal.* **2019**, *9* (4), 3537–3550.
- (9) Subramaniam, S.; Guo, M. F.; Bathena, T.; Gray, M.; Zhang, X.; Martinez, A.; Kovarik, L.; Goulas, K. A.; Ramasamy, K. K. Direct Catalytic Conversion of Ethanol to C₅₊ Ketones: Role of Pd–Zn Alloy on Catalytic Activity and Stability. *Angew. Chem., Int. Ed.* **2020**, *59* (34), 14550–14557.
- (10) Wang, Z.; Yin, M.; Pang, J.; Li, X.; Xing, Y.; Su, Y.; Liu, S.; Liu, X.; Wu, P.; Zheng, M.; Zhang, T. Active and Stable Cu doped NiMgAlO Catalysts for Upgrading Ethanol to n-Butanol. *J. Energy Chem.* **2022**, *72*, 306–317.
- (11) Moteki, T.; Flaherty, D. W. Mechanistic Insight to C–C Bond Formation and Predictive Models for Cascade Reactions among Alcohols on Ca- and Sr-Hydroxyapatites. *ACS Catal.* **2016**, *6* (7), 4170–4183.
- (12) Liu, H.; Zheng, T.; Hui, T.; Zhang, R.; Meng, X.; Zhu, W.; Liu, H.; Liu, Z. Synthesis of n-butanol from Ethanol over Pt-Y/beta Catalyst: Synergistic Catalysis of Yttrium and Platinum Site. *Chem. Eng. J.* **2024**, *481*, No. 148397.
- (13) Yuan, B.; Zhang, J.; An, Z.; Zhu, Y.; Shu, X.; Song, H.; Xiang, X.; Wang, W.; Jing, Y.; Zheng, L.; He, J. Atomic Ru Catalysis for Ethanol Coupling to C₄₊ Alcohols. *Appl. Catal., B* **2022**, *309*, No. 121271.
- (14) Lv, W.-L.; He, L.; Li, W.-C.; Zhou, B.-C.; Lv, S.-P.; Lu, A.-H. Atomically Dispersed Co²⁺ on MgAlOx Boosting C_{4–10} Alcohols Selectivity of Ethanol Valorization. *Green Chem.* **2023**, *25* (7), 2653–2662.
- (15) Jiang, D.; Fang, G.; Tong, Y.; Wu, X.; Wang, Y.; Hong, D.; Leng, W.; Liang, Z.; Tu, P.; Liu, L.; Xu, K.; Ni, J.; Li, X. Multifunctional Pd@UiO-66 Catalysts for Continuous Catalytic Upgrading of Ethanol to n-Butanol. *ACS Catal.* **2018**, *8* (12), 11973–11978.
- (16) Gu, J.; Gong, W.; Zhang, Q.; Long, R.; Ma, J.; Wang, X.; Li, J.; Li, J.; Fan, Y.; Zheng, X.; Qiu, S.; Wang, T.; Xiong, Y. Enabling Direct-growth Route for Highly Efficient Ethanol Upgrading to Long-chain Alcohols in Aqueous Phase. *Nature Commun.* **2023**, *14* (1), No. 7935.
- (17) Brasil, H.; Bittencourt, A. F. B.; Yokoo, K. C. E. S.; Mendes, P. C. D.; Verga, L. G.; Andriani, K. F.; Landers, R.; Da Silva, J. L. F.; Valença, G. P. Synthesis Modification of Hydroxyapatite Surface for Ethanol Conversion: The Role of the Acidic/Basic Sites Ratio. *J. Catal.* **2021**, *404*, 802–813.
- (18) Rodrigues, E. G.; Keller, T. C.; Mitchell, S.; Pérez-Ramírez, J. Hydroxyapatite, an Exceptional Catalyst for the Gas-Phase Deoxygenation of Bio-Oil by Aldol Condensation. *Green Chem.* **2014**, *16* (12), 4870–4874.
- (19) Lu, B.; Ma, S.; Liang, S.; Wang, Z.; Liu, Y.; Mao, S.; Ban, H.; Wang, L.; Wang, Y. Efficient Conversion of Ethanol to 1-Butanol over Adjacent Acid–Base Dual Sites via Enhanced C–H Activation. *ACS Catal.* **2023**, *13* (7), 4866–4872.
- (20) Gu, J.; Mao, K.; Zhang, Q.; Chen, B.; Dong, H.; Qiu, S.; Meng, Q.; Xiong, Y.; Song, J.; Wang, T. N-rich Doping Strategy for Constructing Ni@NC Catalysts to Boost Aqueous Ethanol Coupling towards Higher Alcohols by Inhibiting C₁ Byproducts. *Chem. Eng. J.* **2023**, *453*, No. 139583.
- (21) Scalbert, J.; Thibault-Starzyk, F.; Jacquot, R.; Morvan, D.; Meunier, F. Ethanol Condensation to Butanol at High Temperatures over a Basic Heterogeneous Catalyst: How Relevant is Acetaldehyde Self-Aldolization? *J. Catal.* **2014**, *311*, 28–32.
- (22) Wang, Q.-N.; Zhou, B.-C.; Weng, X.-F.; Lv, S.-P.; Schüth, F.; Lu, A.-H. Hydroxyapatite Nanowires Rich in [Ca–O–P] Sites for Ethanol Direct Coupling Showing High C_{6–12} Alcohol Yield. *Chem. Commun.* **2019**, *55* (70), 10420–10423.
- (23) Eagan, N. M.; Lanci, M. P.; Huber, G. W. Kinetic Modeling of Alcohol Oligomerization over Calcium Hydroxyapatite. *ACS Catal.* **2020**, *10* (5), 2978–2989.
- (24) Tsuchida, T.; Kubo, J.; Yoshioka, T.; Sakuma, S.; Takeguchi, T.; Ueda, W. Reaction of Ethanol over Hydroxyapatite Affected by Ca/P Ratio of Catalyst. *J. Catal.* **2008**, *259* (2), 183–189.
- (25) Ogo, S.; Onda, A.; Iwasa, Y.; Hara, K.; Fukuoka, A.; Yanagisawa, K. 1-Butanol Synthesis from Ethanol over Strontium Phosphate Hydroxyapatite Catalysts with Various Sr/P Ratios. *J. Catal.* **2012**, *296*, 24–30, DOI: 10.1016/j.jcat.2012.08.019.
- (26) Zhou, B.-C.; Li, W.-C.; Lv, W.-L.; Xiang, S.-Y.; Gao, X.-Q.; Lu, A.-H. Enhancing Ethanol Coupling to Produce Higher Alcohols by Tuning H₂ Partial Pressure over a Copper-Hydroxyapatite Catalyst. *ACS Catal.* **2022**, *12* (19), 12045–12054.
- (27) Wang, Q.-N.; Weng, X.-F.; Zhou, B.-C.; Lv, S.-P.; Miao, S.; Zhang, D.; Han, Y.; Scott, S. L.; Schüth, F.; Lu, A.-H. Direct, Selective Production of Aromatic Alcohols from Ethanol Using a Tailored Bifunctional Cobalt–Hydroxyapatite Catalyst. *ACS Catal.* **2019**, *9* (8), 7204–7216.
- (28) Wang, J.; Li, W.-C.; Sun, D.-H.; He, L.; Zhou, B.-C.; Lu, A.-H. High-Selective Upgrading of Ethanol to C_{4–10} Alcohols over Hydroxyapatite Catalyst with Superior Basicity. *ACS Sustainable Chem. Eng.* **2025**, *13* (1), 36–45.
- (29) Chen, W.; Song, X.; Ning, L.; Ding, Y. Ammonia Hydrothermally Treated SiO₂ Supported Rh-Based Catalyst for CO Hydrogenation to C₂ Oxygenates: Remarkable Effect of Support Pore Size. *Ind. Eng. Chem. Res.* **2020**, *59* (42), 18798–18807.
- (30) Xue, M.; Jin, Z.; Yang, B.; Xia, C.; Zhu, G. Boosting Higher Alcohols Selectivity via Regulating Basicity of Ni/Hydroxyapatite in Ethanol Upgrading. *ACS Catal.* **2024**, *14* (16), 12654–12663.
- (31) Kiani, D.; Baltrusaitis, J. Surface Chemistry of Hydroxyapatite for Sustainable n-Butanol Production from Bio-ethanol. *Chem. Catal.* **2021**, *1* (4), 782–801.
- (32) Ho, C. R.; Shylesh, S.; Bell, A. T. Mechanism and Kinetics of Ethanol Coupling to Butanol over Hydroxyapatite. *ACS Catal.* **2016**, *6* (2), 939–948.
- (33) Diallo-Garcia, S.; Osman, M. B.; Krafft, J.-M.; Casale, S.; Thomas, C.; Kubo, J.; Costentin, G. Identification of Surface Basic Sites and Acid–Base Pairs of Hydroxyapatite. *J. Phys. Chem. C* **2014**, *118* (24), 12744–12757.

- (34) Oh, S. C.; Xu, J.; Tran, D. T.; Liu, B.; Liu, D. Effects of Controlled Crystalline Surface of Hydroxyapatite on Methane Oxidation Reactions. *ACS Catal.* **2018**, *8* (5), 4493–4507.
- (35) Ospina, C. A.; Terra, J.; Ramirez, A. J.; Farina, M.; Ellis, D. E.; Rossi, A. M. Experimental Evidence and Structural Modeling of Nonstoichiometric (010) Surfaces Coexisting in Hydroxyapatite Nano-Crystals. *Colloids Surf., B* **2012**, *89*, 15–22.
- (36) Wang, S.; Song, L.; Qu, Z. Cu/ZnAl₂O₄ Catalysts Prepared by Ammonia Evaporation Method: Improving Methanol Selectivity in CO₂ Hydrogenation via Regulation of Metal-Support Interaction. *Chem. Eng. J.* **2023**, 469, No. 144008.
- (37) Posner, A. S.; Stutman, J. M.; Lippincott, E. R. Hydrogen-Bonding in Calcium-Deficient Hydroxyapatites. *Nature* **1960**, *188* (4749), 486–487.
- (38) Wang, S. C.; Cendejas, M. C.; Hermans, I. Insights into Ethanol Coupling over Hydroxyapatite using Modulation Excitation Operando Infrared Spectroscopy. *ChemCatChem.* **2020**, *12* (16), 4167–4175.
- (39) Hanspal, S.; Young, Z. D.; Shou, H.; Davis, R. J. Multiproduct Steady-State Isotopic Transient Kinetic Analysis of the Ethanol Coupling Reaction over Hydroxyapatite and Magnesia. *ACS Catal.* **2015**, *5* (3), 1737–1746.
- (40) Ramesh, K.; Ling, E. G. Y.; Gwie, C. G.; White, T. J.; Borgna, A. Structure and Surface Reactivity of WO₄²⁻, SO₄²⁻, PO₄³⁻ Modified Ca-Hydroxyapatite Catalysts and Their Activity in Ethanol Conversion. *J. Phys. Chem. C* **2012**, *116* (35), 18736–18745.
- (41) Bernhard, Y.; Thomson, B.; Ferey, V.; Sauthier, M. Nickel-Catalyzed α -Allylation of Aldehydes and Tandem Aldol Condensation/Allylation Reaction with Allylic Alcohols. *Angew. Chem., Int. Ed.* **2017**, *56* (26), 7460–7464.